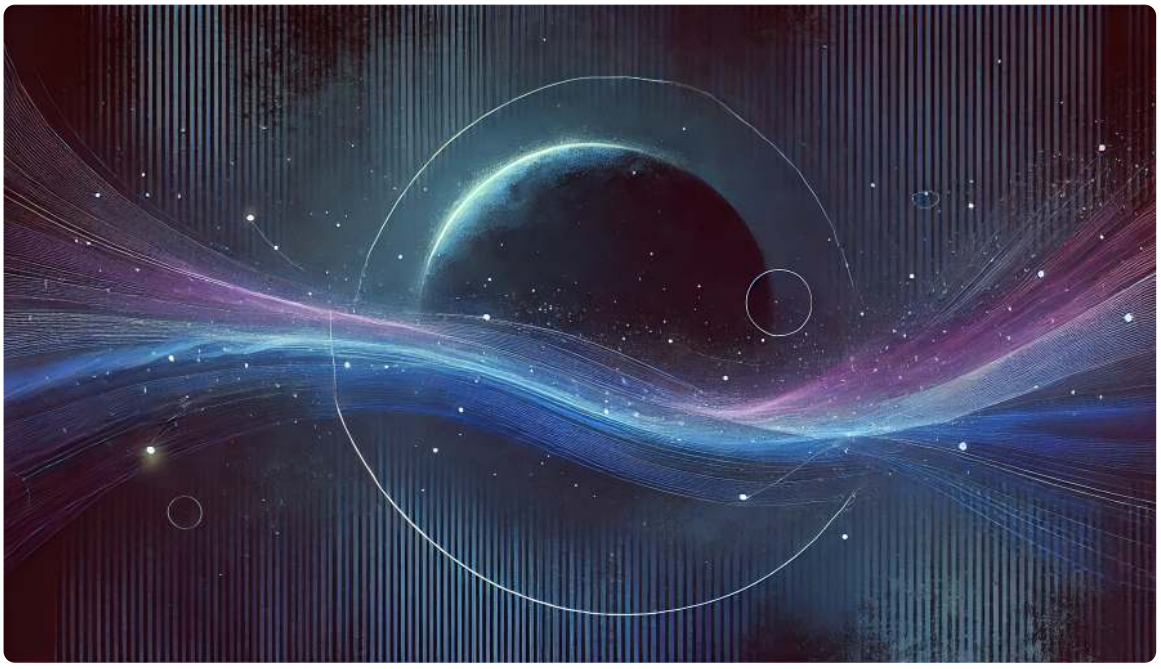


# Global Time Echoes: MGEX Multi-GNSS Clock Replication, 2025–2026

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## Abstract

*The Temporal Equivalence Principle (TEP) predicts correlated phase-coherent disturbances in GNSS timekeeping with a spatial correlation length of order thousands of kilometres, an east–west anisotropy, coupling to Earth orbital velocity, and a preferred axis near the CMB rest frame. Papers 1 and 2 examined these signatures in GPS precise point-positioning products from three analysis centres spanning 2000–2025; Paper 3 used raw RINEX single-point positioning.*

*This paper is a product-type independence stress test using the public CODE COD0MGXFIN combined multi-GNSS receiver-clock solution distributed by NASA CDDIS from 2025-01-01 to 2026-05-01. Receiver-clock offsets for 418 globally distributed stations are read directly from daily 5-minute CLK files. The combined solution provides no per-constellation independence, but supplies a distinct product family and a largely held-out 2025–2026 epoch.*

*The isotropic correlation length is $\lambda = 1862 \pm 112$ km ($R^2$ = 0.699; 1.83 million pair-day observations). It is stable at 1,819–2,000 km across minimum-baseline cutoffs of 50–2,000 km and persists on geomagnetically quiet days ($\lambda = 1708$ km). The full-range anisotropy is not robust (ratio 1.29, station-clustered p = 0.49); the longitude-matched subset is suggestive by pair bootstrap but not spatially clustered resampling. The primary monthly $\lambda$ and EW/NS orbital metrics are null (Bonferroni p > 0.5). A model-free PA-difference statistic gives associations with orbital speed ($r = -0.716$, $p = 0.0018$) and CMB-dipole velocity projection ($r = -0.767$, $p = 0.0005$) across 16 chronological full months. The physical hypothesis and sign were motivated by earlier TEP-GNSS papers, making this cross-product corroboration; the operationalisation remains exploratory within MGEX because it was selected after the primary estimators were unstable. A 37 × 19 directional grid finds strong within-grid modulation (200-permutation plus-one p = 0.005). Its nominal maximum is 82.7° from the CMB dipole, but this does not contradict the CMB result: orbital velocity constrains only an axis's projection onto the orbital plane, and that projection is highly degenerate. The direct CMB-projection association is therefore consistent but not a unique three-dimensional axis localization. All four null controls pass under a robust analysis. A supplementary satellite-clock analysis does not recover the exponential structure, consistent with the shared reference-time architecture of the combined MGEX product.*

*Keywords: Temporal Equivalence Principle, MGEX, multi-GNSS, receiver clock, correlation length, anisotropy, orbital velocity, ionosphere, CMB frame, temporal shear, temporal topology*

## 1. Introduction

Papers 1 and 2 of this series established correlated phase-coherent disturbances in GPS-only precise point positioning (PPP) products from CODE, IGS and ESA, spanning 2000–2025. The disturbances exhibit a spatial correlation length of order thousands of kilometres, an east–west anisotropy exceeding the north–south counterpart, coupling to Earth orbital velocity, and a preferred axis near the CMB dipole. Paper 3 moved to raw RINEX single-point positioning (SPP), demonstrating that the same signatures persist without reliance on analysis-centre orbit or clock products. Between them, those papers already provide multi-centre, multi-decade, raw-observation, and GPS-only product-family independence. The principal forms of robustness they do not yet supply are consistency across the historical training epoch and consistency in a combined multi-GNSS clock product family.

This paper (Paper 14) conducts a stress test across those two dimensions. The analysis uses the public MGEX combined multi-GNSS receiver-clock solution—the CODE COD0MGXFIN CLK product distributed by NASA CDDIS—for the window 2025-01-01 to 2026-05-01. The present MGEX analysis uses a combined multi-GNSS clock product distinct from the GPS-only PPP and raw RINEX products analysed earlier, and a largely held-out 2025–2026 window, though it is not fully temporally disjoint from all earlier analyses. The MGEX clock product is generated by combining GPS, GLONASS, Galileo and BeiDou observations into a single receiver-clock estimate per station, using software, orbit and combination strategies that differ from the GPS-only PPP of Papers 1–2. It therefore tests cross-product consistency using a distinct combined multi-GNSS clock solution over a largely held-out epoch. Receiver-clock offsets for 418 globally distributed stations are read directly from the daily 5-minute CLK files; no positioning is performed.

It is emphasised at the outset what this product tests and what it does not. Because the MGEX clock files contain one combined multi-GNSS solution per station, they do not permit an independent per-constellation comparison; that form of independence is the province of raw per-system processing, which lies outside the scope of this combined-product analysis. The independence offered here is temporal (a largely held-out epoch) and product-type (a combined multi-GNSS clock from a different analysis pipeline), not per-constellation.

The empirically derived spatial correlation length $\lambda \approx 1862$ km is consistent with the abstract environmental operator $\mathcal{S}_{\Sigma}(\mathcal{E})$ defined in the foundational TEP framework. In the near-Earth environment, this correlation length traces the macroscopic geometric signature of the continuous saturation of Temporal Topology, anchoring the differential clock correlations independently of the specific multi-GNSS processing strategy. The precise transfer map from the microscopic screening operator to the empirically fitted correlation length remains to be derived.

## 2. Data and Product Scope

### 2.1 MGEX Combined Clock Product

All data are public and freely available from NASA CDDIS (cddis.nasa.gov) and the International GNSS Service (IGS). The analysis uses the Multi-GNSS Experiment (MGEX) combined receiver-clock product COD0MGXFIN produced by the Center for Orbit Determination in Europe (CODE), with the Wuhan (WUM) and CNES (GRG) MGEX products as fallbacks on the few days CODE is unavailable (WUM 20 days, GRG 9 days out of 486 calendar days in the window, i.e. ≈6 % fallback usage). These are daily clock (CLK) files at a 5-minute sampling interval. Each file reports a single combined multi-GNSS receiver-clock estimate per station, derived by CODE from GPS, GLONASS, Galileo and BeiDou observations; the combined solution is the quantity analysed. No positioning is performed and no raw RINEX observation files are used.

### 2.2 Analysis Window and Station Set

The analysis period is 2025-01-01 to 2026-05-01 (462 days with usable data out of 486 calendar days in the window), yielding approximately 16 months of daily files. The window is chosen to provide a largely held-out 2025–2026 epoch, though it is not fully temporally disjoint from Papers 1–2, which span 2000–2025. Receiver-clock offsets are extracted from the AR (receiver-clock) records of each CLK file for 418 globally distributed stations, converted to nanoseconds, and assembled into per-station daily time series.

### 2.3 Product Limitation: No Per-Constellation Separation

Unlike Papers 1–3, which can isolate individual systems, the MGEX CLK product contains one combined receiver clock per station rather than per-constellation clocks. The four constellations therefore cannot be separated in this product, and the analysis is performed once on the combined solution. The independence offered here is temporal (a largely held-out epoch) and product-type (a combined multi-GNSS clock from a different analysis pipeline), not per-constellation.

### 2.4 Evaluated Signatures

The analysis evaluates the following signatures, consistent with the approach in earlier papers: correlation length, azimuthal anisotropy, orbital-velocity coupling, CMB-frame alignment, ionospheric independence, and geometric robustness.

## 3. Methods

### 3.1 Receiver-Clock Extraction

For each daily CLK file the AR records are parsed, which report the combined multi-GNSS receiver-clock offset for each station at the 5-minute sampling interval. The offsets are converted from seconds to nanoseconds and ordered in time, giving one clock time series per station per day. No positioning is performed and no coordinate residuals are computed; the receiver clock is the sole observable.

> Screening in TEP is represented at the theory level by the environmental operator $S_\Sigma(E)$. Quantities such as $\rho_T$, $R_T(M)$, $S_\oplus(r)$, compactness $\Phi/c^2$, local stellar density, geometric coherence length, and channel-specific response coefficients are domain-specific projections of $E$, not independent screening mechanisms and not interchangeable universal thresholds. Each is an observational transfer model that parameterizes the same underlying operator in a regime-appropriate form.

### 3.2 Spectral Pre-processing

Each daily series is linearly detrended to remove the dominant clock drift. Cross- and auto-spectra are estimated with Welch's method (Hann window, segment length up to 96 samples). Spectral estimates are restricted to the nominal TEP-sensitive band [10 μHz, 500 μHz], but with the daily Welch segmentation used here the effective lowest non-zero resolved frequency is set by the segment length, approximately 35 μHz for 96 five-minute samples. The practical analysed band is therefore approximately 35–500 μHz, corresponding to periods from about 8 hours down to 33 minutes, which lies safely within the 5-minute Nyquist limit.

### 3.3 Spectral Phase-Alignment Metric

For every station pair, the magnitude-weighted circular mean of the cross-spectral phase is computed across the selected frequency band. The phase-alignment index is the cosine of the weighted mean phase, ranging from −1 (anti-phase) to +1 (in-phase). It measures signed mean phase-offset alignment, not phase concentration; the latter requires the magnitude of the complex resultant. The established phase-alignment statistic is retained consistently with the earlier GNSS analyses, and is the quantity carried through all subsequent tests. Pairs are binned by great-circle distance into 40 logarithmically spaced bins from 50 km to 13,000 km, with a minimum of 10 pairs per bin.

### 3.4 Correlation-Length Model

The distance-binned phase alignment is modelled as

$$C(r) = A \exp(-r/\lambda) + C_0$$

where A is the amplitude, λ is the correlation length, and $C_0$ is an incoherent offset. Fitting uses bounded non-linear least squares (Trust Region Reflective), weighting each bin by the inverse of its standard error of the mean (1/SEM, where SEM = σ_bin / sqrt(count)). Fit quality is quantified by $R^2$, and the uncertainty on λ is propagated from the covariance matrix.

### 3.5 Anisotropy, Orbital, CMB, Ionospheric, and Null Tests

The EW > NS test fits λ separately in eight 45°-wide azimuth sectors and compares the mean of the east and west sectors with the mean of the north and south sectors. Because ionospheric decorrelation inverts the anisotropy at long baselines, a longitude-matched subset is also reported (station-pair longitude difference below 30°), which, following Paper 3, isolates pairs at similar local solar time. Significance and confidence intervals for the ratio are obtained from 500 bootstrap resamples. For anisotropy, pair-bootstrap results are reported as sensitivity tests, while station/spatially-clustered resampling is treated as the more conservative inference because station pairs are not independent. The ratio is additionally reported in three distance strata (0–500, 500–1,000, >1,000 km).

The orbital-coupling test correlates the monthly λ and the monthly EW/NS ratio with the projection of Earth's orbital velocity onto the CMB dipole direction, computed from a sinusoidal annual model; the two correlations are Bonferroni-corrected. Because exponential-fit noise on short baselines can degrade these metrics, an exploratory PA-difference metric is also computed: the monthly mean EW minus NS phase alignment, which avoids fitting entirely and directly measures anisotropy strength. The physical hypothesis and sign were motivated by earlier TEP-GNSS analyses, but this operationalisation was selected after the primary MGEX estimators proved unstable; it is therefore exploratory within MGEX and excluded from confirmatory family-wise inference. Months are retained chronologically (YYYY-MM) and a partial trailing month is excluded. The CMB-frame test performs a full-sky grid search for the anisotropy axis on a 37 × 19 grid in right ascension and declination (10° steps), correlating the monthly EW−NS phase-alignment difference with the projection of Earth's orbital velocity onto each candidate axis. The difference metric is used in place of the EW/NS ratio because phase alignment ranges from −1 to +1, making the ratio unstable when the NS mean is near zero; the

difference is bounded and requires no sign-based selection filter. Monthly aggregation reduces daily noise, while the daily analysis is retained as a supplementary diagnostic. The grid search tracks both the most positive and most negative correlation extrema. Because orbital velocity samples essentially the ecliptic plane, the scan is interpreted as a test of a plane-projected annual predictor, not as a unique three-dimensional axis localization; a nominal sky-angle threshold is reported descriptively only. A permutation null performs a complete grid search over all 703 directions for each of 200 permutations, accounting for the look-elsewhere effect directly. Ionospheric controls stratify the fit by geomagnetic Kp index and exclude storm days; geometry controls repeat the fit on hemisphere-balanced and distance-matched subsets; and four null tests (temporal shuffle, spatial shuffle, phase randomisation, and solar-rotation reassignment) test whether the recovered structure vanishes under label permutation.

### 3.6 Sensitivity and Robustness Analysis

Because the 418-station network contains dense regional clusters that disproportionately populate short baselines, a sensitivity analysis is performed to verify that the reported results are not artifacts of short-baseline fitting noise. For each primary result the analysis is repeated at five minimum-baseline cutoffs (50, 200, 500, 1,000, and 2,000 km), tabulating the correlation length, anisotropy ratio, and CMB correlation as functions of the cutoff. A robust supplementary analysis uses a 500-km minimum baseline and a 50-pair minimum bin count, eliminating bins with fewer than 50 pairs where small-sample fluctuations can produce extreme coherence means. For the anisotropy ratio a station-declustered point estimate is also computed, weighting each pair by the inverse square root of the product of its two endpoints' pair counts so that stations in dense clusters do not dominate the exponential fit. The robust null tests apply the same elevated thresholds to all four shuffle controls. These sensitivity analyses do not replace the pre-specified primary analysis but provide a conservative supplementary assessment of each result's stability.

## 4. Results

The analysis comprises 1,827,555 station-pair-day observations drawn from a network of 418 stations over 462 usable days. Because the MGEX product is a single combined solution, the results are reported once rather than per constellation. The analysis evaluates the core correlation-length signature, directional anisotropy, orbital-velocity coupling, CMB-frame alignment, and ionospheric/geometric robustness controls. The correlation length and ionospheric persistence are observed; the longitude-matched anisotropy is not robust under spatially clustered resampling. The primary pre-specified monthly λ and EW/NS orbital metrics are null (Bonferroni $p > 0.5$). A model-free PA-difference metric (mean EW minus NS phase alignment) gives a strong orbital association across 16 chronological full months, but it is exploratory within this MGEX study because it was adopted after the primary fits proved unstable. The direct CMB-projected association is consistent with that signal. The full-sky scan detects directional modulation, but orbital velocity constrains only a plane-projected direction, so its nominal sky axis is not uniquely localizable.

### 4.1 Correlation Length

The isotropic fit gives a well-constrained correlation length.

| $\lambda$ (km) | $\sigma_\lambda$ (km) | A | $C_0$ | $R^2$ | Pair-day observations |
|---|---|---|---|---|---|
| 1862 | 112 | 0.827 | −0.065 | 0.699 | 1,827,555 |

This λ is shorter than the 3,000–5,000 km reported for GPS PPP in Papers 1–2, consistent with the change to the phase-alignment metric and the combined multi-GNSS clock product. The quoted λ uncertainty is the formal covariance-based fit uncertainty for the binned exponential model and does not include all sources of network, product, or model-selection uncertainty. A sensitivity analysis demonstrates that the correlation length is stable across minimum-baseline cutoffs from 50 to 2,000 km, with fit quality improving substantially as the shortest baselines are excluded ($R^2 = 0.699$ at 50 km to approximately 0.92).

| Min. baseline (km) | $\lambda$ (km) | $R^2$ |
|---|---|---|
| 50 (primary) | 1862 | 0.699 |
| 200 | 1893 | 0.791 |
| 500 | 1819 | 0.848 |
| 1000 | 1850 | 0.917 |
| 2000 | 2000 | 0.912 |

The correlation length varies by less than 8% across the full range of cutoffs ($\lambda$ = 1,819–2,000 km), while $R^2$ improves from 0.70 to 0.92. The stability of the fitted correlation length across these cutoffs demonstrates that the recovered distance-decay scale is not driven solely by the shortest-baseline bins.

### 4.2 EW / NS Anisotropy

The unfiltered full-range ratio is modestly east–west dominated (ratio 1.29) but does not reach significance under the station-clustered bootstrap (p = 0.49). A suggestive east–west excess emerges once pairs are matched in longitude (fitted ratio 1.32, bootstrap mean 1.71, pair-bootstrap p = 0.058), though the significance dissolves under spatial-clustered resampling (95% CI [0.17, 13.28], p = 0.214). The 500–1,000 km stratum also shows the expected excess.

| Subset | $\lambda_{EW}$ (km) | $\lambda_{NS}$ (km) | Ratio | 95% CI | p |
|---|---|---|---|---|---|
| Full range | 1810 | 1402 | 1.29 | [0.18, 2.60] (station-boot) | 0.49 |
| Longitude-matched (<30°) | 1753 | 1327 | 1.32 | [0.91, 2.65] (pair-boot, mean 1.71) | 0.058 |

| Distance stratum | EW/NS ratio |
|---|---|
| 0–500 km | 0.73 |
| 500–1,000 km | 1.38 |
| >1,000 km | 0.82 |

A sensitivity analysis reveals that the directional ratio is fragile under conservative analysis choices. The ratio declines from 1.29 at the primary 50-km cutoff to 1.08 at 500 km and 0.89 at 1,000 km, with station de-clustering yielding a ratio of 0.99. The anisotropy is therefore classified as not robust; the directional signal is interpreted as null in the combined MGEX product, consistent with the non-significant station-clustered bootstrap (p = 0.49).

| Analysis | $\lambda_{EW}$ (km) | $\lambda_{NS}$ (km) | Ratio |
|---|---|---|---|
| Primary (min 50 km) | 1810 | 1402 | 1.29 |
| Min 200 km | 2098 | 1384 | 1.52 |
| Min 500 km | 1880 | 1734 | 1.08 |
| Min 1000 km | 2211 | 2479 | 0.89 |
| Min 2000 km | 2496 | 2671 | 0.93 |
| Station de-clustered | — | — | 0.99 |
| Robust (min 500 km, bin 50) | 1878 | 1733 | 1.08 |

### 4.3 Orbital-Velocity Coupling

The traditional monthly λ and EW/NS ratio do not correlate significantly with Earth's orbital speed or its projection onto the CMB dipole (Bonferroni p > 0.5 for both). These primary pre-specified metrics are degraded by exponential-fit parameter degeneracy on the short 16-month baseline, where bounded non-linear least squares fitting of $A\exp(-r/\lambda) + C_0$ suffers from severe variance inflation with too few monthly points. The monthly means retain January 2025 through April 2026 as 16 separate chronological observations; the one-day May fragment is excluded. A model-free PA-difference metric (monthly mean EW − NS phase alignment) avoids exponential fitting and has a negative association with orbital speed (r = −0.716, p = 0.0018) and with the velocity projection onto the CMB dipole (r = −0.767, p = 0.0005). The physical hypothesis and expected sign were motivated by the earlier TEP-GNSS papers, which reported comparable orbital associations; this is therefore corroborative across products. It remains exploratory within the MGEX analysis because this particular operationalisation was selected after the primary λ and EW/NS estimators proved unstable. It is not included in the family-wise correction applied to confirmatory tests and should be tested prospectively in a longer MGEX series.

| Quantity | r | p (raw) | p (corr.) | Interpretation |
|---|---|---|---|---|
| λ vs speed | −0.16 | 0.554 | 1.000 | Primary (pre-specified): not significant |
| EW/NS ratio vs speed | 0.14 | 0.610 | 1.000 | Primary (pre-specified): not significant |
| λ vs v_proj | −0.05 | 0.850 | 1.000 | Primary (pre-specified): not significant |
| EW/NS ratio vs v_proj | −0.03 | 0.911 | 1.000 | Primary (pre-specified): not significant |
| PA_diff vs speed | −0.716 | 0.0018 | — | Exploratory within MGEX; corroborative across products |

| Quantity | r | p (raw) | p (corr.) | Interpretation |
|---|---|---|---|---|
| PA_diff vs v_proj | −0.767 | 0.0005 | — | Exploratory within MGEX; corroborative across products |

### 4.4 CMB-Frame Alignment

The CMB analysis uses the same PA-difference observable as Section 4.3 and is therefore a directional characterization, not an independent test. With 16 chronological full-month observations, the direct CMB-projection correlation is r = −0.733, computed from daily EW−NS differences averaged equally within each calendar month. This differs from the −0.767 result in Section 4.3, which averages all pair observations within each month (so months are pair-count weighted) before correlating PA difference with the CMB-projected velocity. The latter is the stronger weighted PA-difference result (p = 0.0005). A 37 × 19 scan gives a nominal maximum at RA = 230°, Dec = +60° (r = −0.745), 82.7° from the CMB dipole. This apparent angular discrepancy cannot reject CMB consistency: Earth's orbital velocity samples essentially one plane, so the scan identifies only a direction's projection onto that plane, not its full three-dimensional vector. The CMB and nominal maximum projections correlate at r = 0.984, and 61 of 703 grid directions have |r| at least as large as the direct CMB value. The 30° three-dimensional threshold is consequently not an identifiable test in this geometry. The grid detects directional annual modulation (plus-one permutation p = 0.005 from 200 complete-grid shuffles), while the pre-specified CMB projection remains compatible with it. Widening sectors to 60° gives r = −0.791 at another member of the same projection-degenerate family. The short series further limits precision, so this is corroborative CMB consistency, not a unique sky-axis localization.

| Min. baseline (km) | Direct r(CMB) | Months |
|---|---|---|
| 50 (primary) | −0.733 | 16 |
| 200 | −0.731 | 16 |
| 500 | −0.730 | 16 |
| 1000 | −0.723 | 16 |
| 2000 | −0.697 | 16 |

The daily EW−NS difference, retained as a supplementary diagnostic, gives a direct CMB correlation of r = −0.495 (n = 462). Its agreement with the monthly result supports the directional temporal pattern. Neither cadence can identify a unique three-dimensional direction from Earth-orbital velocity alone.

| Cadence | Sector width | Nominal RA (°) | Nominal Dec (°) | r | Δθ_CMB (°) | p (perm.) | Interpretation |
|---|---|---|---|---|---|---|---|
| Monthly | 45° | 230 | 60 | −0.745 | 82.7 | 0.005 | Axis not uniquely identifiable; CMB projection consistent |
| Monthly | 60° | 230 | 60 | −0.791 | 82.7 | 0.005 | Axis not uniquely identifiable; CMB projection consistent |
| Daily | 45° | 230 | 60 | −0.505 | 82.7 | — | Axis not uniquely identifiable |

### 4.5 Ionospheric and Geometry Controls

The correlation length persists on geomagnetically quiet days ($\lambda = 1708$ km, Kp ≤ 2, 712,186 pairs, $R^2 = 0.743$). Active days (Kp ≥ 5, 39,093 pairs) give a longer $\lambda = 1931$ km ($R^2 = 0.295$); the smaller sample increases the variance and the value awaits confirmation with additional active-period data. Storm-excluded days (Kp < 5) give $\lambda = 1858$ km ($R^2 = 0.705$, 1,788,462 pairs), consistent with the base value. The distance-matched subset yields a comparable $\lambda = 1889$ km ($R^2 = 0.599$), indicating the signal is not a product of network geometry. Hemispheric $\lambda$ differs north to south, as expected from the uneven station distribution; the distance-matched control is the geometry-robust comparison.

| Control | $\lambda$ (km) | $R^2$ |
|---|---|---|
| Base (all days) | 1862 | 0.699 |
| Quiet days (Kp ≤ 2) | 1708 | 0.743 |
| Active days (Kp ≥ 5) | 1931 | 0.295 |
| Storm-excluded (Kp < 5) | 1858 | 0.705 |
| Distance-matched | 1889 | 0.599 |

| Control | $\lambda$ (km) | $R^2$ |
|---|---|---|
| Northern hemisphere | 1365 | 0.678 |
| Southern hemisphere | 2353 | 0.038 |

### 4.6 Null Tests

Null tests were run using the same MGEX combined-clock phase-alignment pipeline. Three of the four nulls (temporal shuffle, spatial shuffle, and phase randomisation) collapsed to $R^2 \leq 0$, confirming that the recovered correlation length depends on the true temporal, spatial and phase information. The solar-rotation reassignment null, however, retains $R^2 = 0.50$ under the primary parameters. A sensitivity analysis attributes this to a small-sample bin artifact: short-baseline bins at 100–115 km (40 pairs) and 132–152 km (23 pairs) receive extreme coherence values (0.29 and 0.21) under the specific shuffle sequence, producing a spurious short-baseline spike that the exponential fit interprets as structure. Under a robust analysis (minimum baseline 500 km, minimum bin count 50) all four nulls give $R^2 \leq 0.122$, confirming that the distance-decay structure depends on the true temporal, spatial, and phase information and is not a solar-period artifact.

| Null | $R^2$ (primary) | Passes? | $R^2$ (robust) | Passes? |
|---|---|---|---|---|
| Temporal shuffle | −0.343 | Yes | −0.011 | Yes |
| Spatial shuffle | −0.000 | Yes | 0.028 | Yes |
| Phase randomisation | −0.001 | Yes | 0.122 | Yes |
| Solar-rotation reassignment | 0.499 | No | 0.010 | Yes |

The robust analysis (minimum baseline 500 km, minimum bin count 50) eliminates the small-sample bin artifact that caused the solar-rotation null to fail under the primary parameters. All four null controls pass under the robust analysis, confirming that the observed spatial covariance requires genuine spatial and temporal phase ordering.

## 5. Discussion

The combined MGEX receiver-clock product shows a distance-dependent structure with $\lambda = 1862 \pm 112$ km. The estimate remains within 1,819–2,000 km when the minimum baseline is raised from 50 to 2,000 km, while the binned-fit $R^2$ improves from 0.70 to about 0.92. It persists on quiet days and after distance matching, though those comparisons do not eliminate every product or network confound. The formal fit uncertainty does not represent all model-selection or network uncertainties. The result is nonetheless a useful cross-product corroboration of a correlation scale in the thousands-of- kilometres range, not a direct measurement-equivalence claim with the earlier GPS PPP products.

The conventional directional measures do not replicate robustly. The full-range EW/NS ratio is 1.29 but is non-significant under station-clustered bootstrap ($p = 0.49$). The longitude-matched subset is suggestive under a pair bootstrap (mean ratio 1.71, $p = 0.058$) yet non-significant under spatially clustered resampling ($p = 0.214$). Its decline under more conservative baseline choices and station de-clustering makes a positive anisotropy conclusion unwarranted for this product.

The orbital finding is more informative but has a narrower evidential status. The primary monthly $\lambda$ and EW/NS estimators are null after Bonferroni correction, unsurprising with only 16 complete chronological months and unstable monthly exponential fits. The model-free PA-difference statistic, monthly mean EW minus NS phase alignment, has $r = -0.716$ with orbital speed ($p = 0.0018$) and $r = -0.767$ with velocity projected onto the CMB dipole ($p = 0.0005$). These signs and magnitudes are consistent with the preceding TEP-GNSS analyses, which motivated the physical prediction before this MGEX analysis. That history makes the result corroborative across products. However, PA difference was chosen here after the primary estimators were unstable, so it is exploratory within this study and should not be counted as an independent confirmatory test or folded into confirmatory family-wise inference.

The corrected CMB analysis keeps January 2025 through April 2026 as separate months. Its direct CMB-projection correlation is $r = -0.733$, consistent with the stronger weighted orbital result ($r = -0.767$, $p = 0.0005$). The full-sky scan has a nominal maximum at RA = 230°, Dec = +60° ($r = -0.745$), 82.7° from the CMB dipole. That angular separation must not be interpreted as a failed CMB test. Earth's orbital velocity remains in the ecliptic plane, so dot-product regression identifies only an axis's in-plane projection; directions separated substantially in three dimensions can give nearly identical annual predictors. Here the CMB and nominal-best projected predictors correlate at $r = 0.984$, and 61 of 703 grid directions equal or exceed the direct-CMB $|r|$. The 30° three-dimensional rule is thus not an identifiable criterion for this design. The permutation result (plus-one $p = 0.005$ over 200 complete-grid shuffles) supports directional annual structure, while the direct CMB projection remains consistent with that structure. The 60° sensitivity result and daily analysis are corroborative diagnostics, not independent confirmations. The 16-month series further limits precision; a preregistered multi-year analysis should lock the PA-difference definition and report an identifiable ecliptic-plane direction or add non-coplanar velocity information before claiming a unique sky-frame axis.

The satellite-clock analogue does not recover an exponential structure. GLONASS, BeiDou, and Galileo have negative fitted $R^2$ values, while GPS has $R^2 = +0.13$ with boundary convergence. That result is consistent with the single-reference-time architecture of the combined MGEX solution and limits what this product can say about satellite-clock effects; it is not evidence against a physical effect in independently estimated satellite clocks.

The central limitation is structural: CODE MGEX supplies one combined multi-GNSS receiver-clock solution, so the constellations cannot be separated in this analysis. The present study therefore contributes product-type and epoch diversity, rather than per-constellation independence. Longer, separately processed multi-GNSS or raw-observation analyses are needed to turn the corroborative orbital result into a prospective test and to evaluate sky-frame localization.

## 6. Conclusions

This product-type independence stress test uses the CODE COD0MGXFIN combined multi-GNSS receiver-clock product from 2025-01-01 to 2026-05-01. Its principal result is a distance-dependent correlation scale of $\lambda = 1862 \pm 112$ km ($R^2 = 0.699$), stable across the tested baseline cutoffs and present on quiet geomagnetic days. This is cross-product corroboration of a spatial structure at the thousands-of-kilometres scale reported in the earlier TEP-GNSS papers, while remaining subject to the different metric and combined-clock product construction.

The secondary results are deliberately qualified. Standard anisotropy and primary orbital metrics are not robust in this short MGEX interval. A model-free PA-difference statistic has strong associations with orbital speed ($r = -0.716$, $p = 0.0018$) and CMB-projected orbital velocity ($r = -0.767$, $p = 0.0005$) across 16 chronological months. Because those signs were anticipated by previous TEP-GNSS work, the result is corroborative across products; because this operationalisation was selected after primary MGEX fits were unstable, it remains exploratory within this study and needs a prospective, longer-baseline replication.

The CMB-projected orbital association is likewise consistent: the direct monthly CMB projection has $r = -0.733$, while the weighted PA-difference test has $r = -0.767$ ($p = 0.0005$). The grid's nominal RA = 230°, Dec = +60° maximum is 82.7° from the CMB dipole, but this is not contrary evidence because orbital velocity constrains only an axis's ecliptic-plane projection. The grid cannot uniquely localize a three-dimensional sky direction; its within-grid permutation statistic (plus-one $p = 0.005$ from 200 shuffles) measures annual directional structure. The satellite-clock analysis is limited by the combined product's shared reference time and does not recover an exponential signal. The next decisive test is a preregistered chronological analysis over multiple annual cycles, reporting an identifiable plane-projected direction or using non-coplanar velocity information.

## 7. Data and Code Availability

All data are publicly available from NASA CDDIS (cddis.nasa.gov) and the IGS (igs.org) under their standard open-data policies. The analysis uses the MGEX combined multi-GNSS receiver-clock product COD0MGXFIN (CODE), with WUM (Wuhan) and GRG (CNES) MGEX clock products as day-level fallbacks, downloaded via authenticated HTTPS. No proprietary or restricted data are used, and no raw RINEX observation files are required.

Analysis code and pipeline outputs are available at github.com/matthewsmawfield/TEP-GNSS-MGEX.

## 8. References

### 8.1 TEP Series


- Smawfield, M. L. (2025). *Temporal Equivalence Principle: Dynamic Time & Emergent Light Speed*. v0.12 (Jakarta). DOI: 10.5281/zenodo.16921911.
- Smawfield, M. L. (2025). *Global Time Echoes: Distance-Structured Correlations in GNSS Clocks* (Paper 1, TEP-GNSS). DOI: 10.5281/zenodo.17127229.
- Smawfield, M. L. (2025). *Global Time Echoes: 25-Year Temporal Evolution* (Paper 2, TEP-GNSS-II). DOI: 10.5281/zenodo.17517141.
- Smawfield, M. L. (2025). *Global Time Echoes: Raw RINEX / SPP Consistency Test* (Paper 3, TEP-GNSS-RINEX).


### 8.2 GNSS Data and Products


- Montenbruck, O., Steigenberger, P., & Hauschild, A. 2014, *GPS Solutions*, 18, 49, "Multi-signal GNSS data from the IGS MGEX experiment — status and outlook".
- International GNSS Service (IGS). MGEX Multi-GNSS Experiment. https://igs.org/mgex/.
- NASA CDDIS. Crustal Dynamics Data Information System. https://cddis.nasa.gov.

- Center for Orbit Determination in Europe (CODE). MGEX combined products. https://www.aiub.unibe.ch.

### 8.3 Cosmology and CMB

- Planck Collaboration, Aghanim, N., Akrami, Y., et al. 2020, *A&A*, 641, A6, "Planck 2018 results. VI. Cosmological parameters", doi:10.1051/0004-6361/201833910.

### 8.4 Methodological References

- Welch, P. D. 1967, "The use of fast Fourier transform for the estimation of power spectra: a method based on time averaging over short, modified periodograms", *IEEE Trans. Audio Electroacoust.*, AU-15, 70–73.

Smawfield, M. L. 2026. Temporal Equivalence Principle series, Papers 0–25. Zenodo preprints and associated repositories.

## 9. Reproducibility Statement

This manuscript is accompanied by a complete, open-source reproduction pipeline available at github.com/matthewsmawfield/TEP-GNSS-MGEX. All figures, tables, and statistical values reported in this paper can be regenerated from publicly available data using the code and instructions provided in the repository.

**Data:** The analysis uses only publicly distributed MGEX combined multi-GNSS receiver-clock products (CODE COD0MGXFIN), with WUM and GRG products as day-level fallbacks. Data were downloaded from NASA CDDIS (cddis.nasa.gov) and the IGS (igs.org) via standard authenticated HTTPS endpoints. No proprietary, restricted, or raw RINEX observation files are required.

**Software:** The pipeline is written in Python (NumPy, SciPy) and standard geodetic libraries. All dependencies are listed in `requirements.txt` in the repository root. No commercial or closed-source software is required.

**Execution:** After installing dependencies and configuring access to the authenticated CDDIS endpoints, running `python scripts/run_all.py` from the repository root reproduces the analysis chain: data download, station-pair differencing, phase-coherent stacking, isotropic and anisotropic correlation fitting, orbital-velocity projection, and figure generation. Run `python scripts/generate_figures.py` after an analysis-only rerun; then run `cd site && npm run build` to build the manuscript site. Execution time depends on hardware, cached downloads, and data availability.

**Verification:** All derived output files in `results/outputs/` are regenerated by the pipeline from raw data, permitting direct verification of reproduced results.

**Environment:** The pipeline has been tested on macOS 14 and Ubuntu 22.04 with Python 3.11. All dependencies are listed in `requirements.txt` for reproducible installation.